

# A Pipeline Taxonomy of User Cognitive Biases

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We organize *user* cognitive biases (excluding distortions induced by the algorithm) by the stage of the information pipeline at which they act: *input*, *process*, and *output*. The *process* stage is further divided into three causally ordered sub-steps: *perception* (forming a read on the received signal and the state of the world), *weighting* (deciding how much the signal counts given its source), and *integration* (revising the opinion). A bias is classified by where it is injected in this pipeline rather than by a hypothesized cognitive origin.

## 1 Input

*Distorts which signals reach the agent. For user biases this reduces to two acts of selection.*

- **Post selection (cherry picking / selective exposure)** — the agent chooses which individual items to consume, preferring those that confirm existing belief. A fast, per-item act.
- **Source selection** — the agent chooses whom to follow or attend to, a slow, structural act. Homophily (selecting on similarity) source selection may also run on prestige or novelty.

## 2 Process

*Turns a received signal into an opinion change. We define a normative baseline and treat each bias as a named, signed deformation of one node in that baseline. The baseline computes four weighting cues in parallel—internal compatibility, source value, frequency, and salience—each set to its normatively correct value, and then performs a single Bayesian integration of the surviving weighted signal. Two cross-links are noted where one cue is computed from another; these are treated as findings rather than exceptions.*

### 2.1 Weighting cues (parallel)

**Internal compatibility.** *Baseline: the prior is used as a Bayesian would use it—informative, but not over-weighted. Deformations over-weight the prior, in increasing intensity.*

- **Confirmation bias** — the weight on a signal increases with its agreement with the prior; the parent deformation of this node.
- **Conservatism (Bayesian under-updating)** — the prior is over-weighted, so the opinion moves too little given the evidence.
- **Belief perseverance** — the prior retains its weight even after the evidence that produced it is removed.
- **Backfire effect** — the extreme deformation: weight on contradicting evidence turns negative, reinforcing the prior; an edge phenomenon with limited replication.
- **Cognitive dissonance** — the compatibility check actively rejects incompatible input to preserve coherence.

**Source value.** *Baseline: a source is weighted by its actual reliability. Deformations set this weight from cues other than reliability.*

- **Authority bias** — weight set by the source’s authority.
- **Halo effect** — weight set by an unrelated positive trait; “source is good → source is right.”
- **Ingroup bias** — higher weight for ingroup sources.
- **Outgroup devaluation** — lower weight for outgroup sources.
- **Truth bias** — the default source weight is set too high; what is heard is assumed true.
- **Bandwagon effect** — *cross-link*: source value is computed from perceived prevalence rather than reliability.
- **Familiarity / mere exposure** — *cross-link*: source value is computed from frequency of exposure rather than reliability.

**Frequency / familiarity.** *Baseline: repetition of the same claim carries no extra evidential value; hearing one rumour twice is not two independent observations. Deformations let repetition or ease of recall stand in for evidence.*

- **Illusory truth effect** — a repeated claim is judged truer through fluency, independently of its content.
- **Availability heuristic** — ease of recall stands in for frequency or probability.

**Salience / affect.** *Baseline: the intrinsic valence or arousal of a signal carries no extra weight beyond its evidential content. Deformations let emotional content set the weight.*

- **Negativity bias** — negative information is weighted more, opposing truth bias as a default prior.
- **Saliency / emotional weighting** — emotionally charged information is weighted more.

## 2.2 Integration (sequential)

*Baseline: the surviving weighted signal revises the opinion on its actual logical merit. Deformations distort the update itself, after weighting.*

- **Belief bias** — reasoning effort is applied asymmetrically: flaws are sought in disbelieved conclusions.
- **Semmelweis reflex** — the update is rejected when it threatens a paradigm or core value.
- **Framing** — the update depends on the presentation of the signal rather than its content.
- **Anchoring** — the update over-relies on the first signal received (primacy).
- **Recency** — the update over-relies on the most recent signal.
- **Asymmetric / optimistic updating** — favourable signals move the opinion more than unfavourable ones.

## 3 Output

*Distorts the expressed opinion relative to the internal one.*

- **Courtesy bias** — a more socially acceptable opinion is reported than the one held.
- **Edgellord effect** — a more extreme opinion is reported than the one held, to gain attention.
- **Group shift / polarization** — group interaction pushes expressed positions to be more extreme than private ones.
- **Spiral of silence** — agents who perceive themselves to be in the minority self-censor.
- **Group think** — dissent is suppressed to preserve harmony.
- **Saying-is-believing** — the act of expressing an opinion shifts the internal one, feeding back into the process stage.

## 4 Miscellanea

*Items set aside from the main taxonomy, retained here so nothing is lost. Grouped by the reason for exclusion rather than by pipeline stage.*

**Excluded as algorithm-induced (deferred to the algorithm layer).** *These distort the input stream but originate in the platform's ranking, not in the user. They belong to the selection layer to be modelled separately.*

- **Engagement / narrative bias** — ranking by highest or lowest opinion to maximise engagement or push a narrative.
- **Filter bubble (imposed)** — ranking by closest opinion, producing homophilic exposure the user did not choose. Distinct from user-driven source selection, which produces the same effect organically.

Reclassified as emergent (observed, not primitive).

- **Shared information bias** — groups over-circulate already-known information. The individual driver is an output tendency (expressing shared content more readily); the over-circulation is the group-level consequence.
- **Echo chamber / polarization** — collective outcomes to be measured as results of the model, not encoded as agent rules.
- **Pluralistic ignorance (as outcome)** — listed under perception as a mechanism, but its population-level signature is itself an emergent phenomenon worth measuring.

Dropped for poor fit (no clean single node).

- **Choice-supportive bias / post-purchase rationalisation** — post-commitment upweighting of a chosen option. Sits awkwardly between integration and output; it is an identity/commitment effect rather than a per-signal weighting. Recoverable as its own coherence node near output if needed.
- **Saying-is-believing** — retained in output, but flagged separately as the one backward arrow (output feeding back into process); it breaks strict feed-forward and may need explicit treatment.

Merged or renamed (kept under another name).

- **Courtesy bias** — originally listed twice (group context and communication); merged into a single output item.
- **Conservatism (original sense)** — the user’s original “conservatism bias” described non-collapse of a logical chain; split into *belief perseverance* (chain non-collapse) and *conservatism / Bayesian under-updating* (moving too little), which are distinct.
- **Homophily** — kept as the dominant criterion of *source selection* rather than as a standalone bias.

Boundary cases (placement still open).

- **False consensus / majority illusion** — placed under perception, but sit on the input/perception boundary; they migrate toward input if perception is defined as “estimating the global state” rather than “interpreting the received signal.”
- **Simmelweis reflex** — placed under integration as paradigm-level rejection, but defensibly belongs under internal compatibility next to confirmation bias.
- **Bandwagon, familiarity / mere exposure** — kept as source-value items but tagged as cross-links (one weighting cue computed from another); could alternatively be elevated to standalone findings.

**Candidate additions not yet placed.** *Mechanisms raised in discussion that are not in the current list but may warrant inclusion.*

- **Emotional contagion** — affect spreading between agents independently of content.
- **Affective polarization** — distance measured on identity/affect rather than on issue position; increasingly the quantity polarization research targets.
- **Primacy/recency as a pair** — recency added as anchoring's twin; confirm both are wanted once sequential feeds are modelled.